



Research Proposal

Department of Physical Science, University of Vauniya

High Latency in Starlink Makes Competitive Gaming Impossible in Rural Areas

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Research Problem & Motivation

The Problem

This research investigates whether Starlink latency characteristics—including periodic spikes caused by satellite handovers—render competitive online gaming unviable for players in rural areas who depend on satellite connectivity.

Competitive gaming requires **sub-50 ms latency** for fair play. Rural communities face a persistent broadband divide—21% of rural Americans lack fixed broadband. Starlink (25–60 ms typical latency) has become the primary solution, but documented 15-second satellite reconfiguration intervals cause periodic latency spikes with direct equity implications for esports participation.

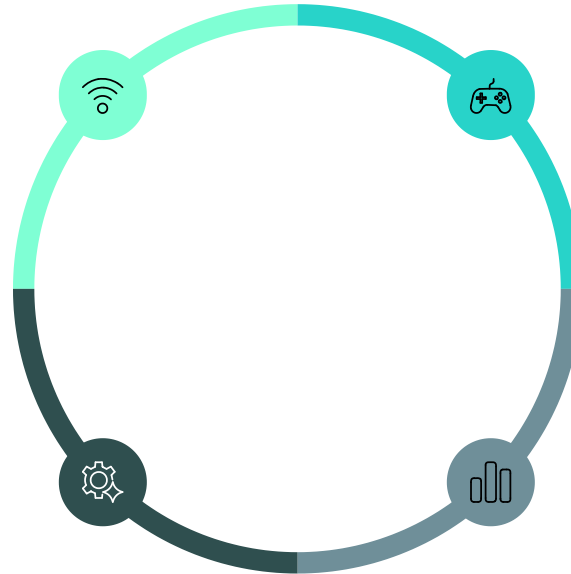
Aim, Objectives & Research Questions

Objective 1

Measure Starlink latency, jitter, and packet loss during competitive gaming in rural locations.

Objective 4

Evaluate whether handover-aware compensation techniques can reduce the competitive disadvantage.



Objective 2

Quantify the impact of satellite handover-induced latency spikes on player performance in FPS and MOBA titles.

Objective 3

Compare Starlink gaming performance against fiber and 5G baselines to establish a competitive gap.

RQ1

What are the latency distributions, jitter, and packet loss during competitive gaming over Starlink in rural areas?

RQ2

How do 15-second satellite handovers impact competitive gaming performance (accuracy, K/D ratio, win rate)?

RQ3

Does Starlink latency create a statistically significant competitive disadvantage vs. fiber/cable players?

RQ4

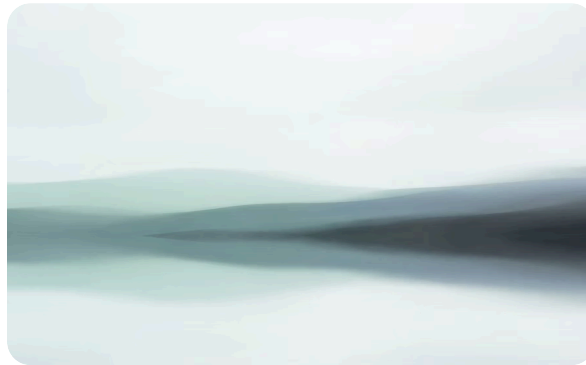
Can handover-aware compensation techniques reduce the performance gap?

Key Findings from Prior Work



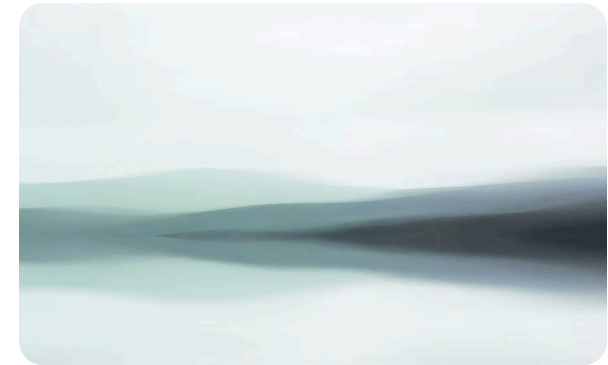
Starlink Performance

Starlink achieves 25–60 ms latency at ~550 km altitude. Mohan et al. identified globally synchronized **15-second reconfiguration intervals** causing latency spikes from 19.2M speed tests across 34 countries. Tiwari et al. characterized tail latencies, developing predictors improving performance by 25%.



Cloud Gaming over Starlink

Tolouei tested Rocket League on GeForce NOW over Starlink, finding handovers at predictable intervals caused input latency spikes, packet loss, and stutters. While playable, these periodic disruptions are **problematic for competitive scenarios**.

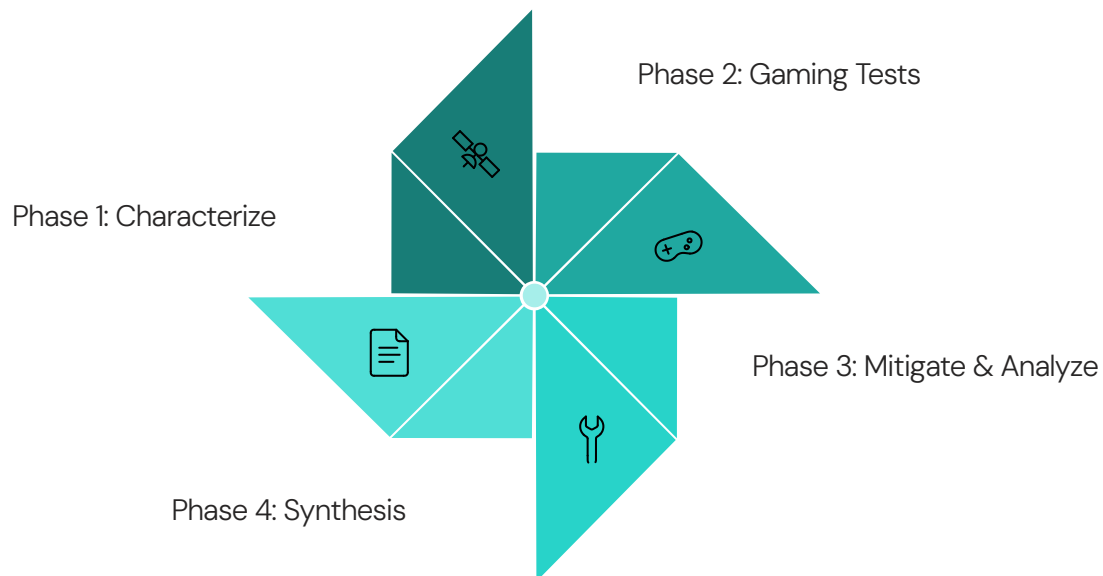


Gaming Thresholds & Rural Divide

Raaen & Grønli established FPS games require **< 50 ms for competitive play**. Olson et al. documented rural residents are less likely to engage in online gaming. Legg argued "latency is becoming the new currency" of the digital divide.

⚠ Research Gap: No controlled user study measures Starlink's impact on competitive gaming performance across rural locations. The effect of 15-second handover spikes on specific in-game events is unquantified, and no formal competitive equity analysis exists.

Proposed Methodology



Tools: Wireshark, tcpdump, Python scripts, game replay APIs, R/SPSS, PyTorch for predictive handover model.



Phase 1 – Network Characterization

ICMP ping at 10 ms intervals, UDP game-traffic emulation, capturing RTT distributions, jitter, packet loss, and handover events across 3 rural sites co-located with fiber baselines.



Phase 2 – Controlled Gaming Experiments

40+ participants (Starlink rural vs. fiber baseline), matched by skill. Counter-Strike 2 (FPS) and MOBA sessions – 20+ matches each – measuring accuracy, K/D, reaction time, win rate, and QoE surveys.



Phase 3 – Mitigation & Analysis

Handover-aware game traffic optimizer exploiting the predictable 15-second schedule: pre-buffer state updates, delay non-essential packets during handover windows, apply client-side input prediction during spikes. Implemented as a local proxy.

OUTCOMES

Expected Outcomes



Starlink Latency Profile

Comprehensive Starlink latency profile during competitive gaming, including handover-correlated spike analysis and 99th-percentile characterization.



Quantified Competitive Disadvantage

Measured accuracy, K/D, and win-rate degradation for Starlink vs. terrestrial players, plus handover-gameplay correlation maps showing how 15-second cycles align with performance drops.



Policy & Publications

Validated handover-aware optimizer, policy recommendations for esports organizations, game developers, and broadband policymakers. Target: 2–3 publications at ACM CHI, ACM MMSys, or IEEE NetGames.

TIMELINE

Research Timeline



KEYWORDS

Thank You

Questions & Discussion

Keywords

Starlink · Latency · Competitive Gaming · Rural Broadband · Esports · LEO Satellite · Jitter · Digital Divide



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